Terms of Reference

Working Group 5: Amplify the Impact of Cryospheric Science

The Decade of Action for Cryospheric Sciences (2025-2034)

Background

On August 13, 2024, the United Nations General Assembly adopted Resolution 78/321, proclaiming 2025 to 2034 as the Decade of Action for Cryospheric Sciences (DACS). The resolution aims to accelerate efforts to address the urgent challenges posed by the melting cryosphere through enhanced scientific research, monitoring, and international cooperation. The resolution highlights the need to understand the cryosphere's role in the Earth's climate system, freshwater resources, ecosystems, and human social, political, and economic activities. It also calls for raising awareness of cryospheric loss and encourages the sharing of best practices to build resilience against related risks. Member States, UN bodies, and stakeholders are invited to support these activities. UNESCO is tasked with leading implementation of the Decade, with other UN organizations and partners, using voluntary contributions to develop programmes.

During the first year of the Decade, 2025, an Ad Hoc Committee was formed by UNESCO to form the structures necessary to implement the Decade. These are summarised in a **Synthesis Document**, which outlines the vision for the Decade. As a result of this process, the committee identified five Overarching Challenges (OCs) for the Decade:

OC1: Global Cryosphere Monitoring
OC2: Actionable Cryosphere Projections
OC3: Cryospheric Risk Management Solutions
OC4: Adaptation to Cryosphere Loss
OC5: Amplify the Impact of Cryospheric Science

Within the Decade, Working Groups aligned with each Overarching Challenge serve as the *main engines of delivery*. The role of Working Group 5 is to achieve the Main Goals (MGs) of **Overarching Challenge 5, Amplify the Impact of Cryospheric Science**:

MG5.1: Policy Integration and Engagement - Cryosphere Loss Prevention
MG5.2: Global Cryosphere Outreach and Communication
MG5.3: Education and Capacity Building

The Decade is split into three triennia. Each MG is assigned a 3-year block, marked by a triennial summit. It is anticipated that appointment to each working group will be on an initial three-year basis, with membership rotating as focus shifts towards subsequent MGs.

Over the course of the Decade, this Working Group is to make cryospheric science a clear, credible, and unavoidable driver of climate decision-making, greenhouse gas emissions reductions, including the accelerated phase-down of fossil fuel use, and long-term action through authoritative evidence and ethical clarity.

Objectives

Rapid and irreversible changes in the cryosphere represent one of the clearest indicators of climate system destabilisation and a critical driver of global climate risk. While the scientific evidence is robust and well established, it is not consistently translated into timely, effective, and proportionate policy responses.

OC5 seeks to strengthen the interface between cryospheric science and decision-making at global, national, and societal levels. The objective of the Working Group 5 is to **amplify the influence of cryospheric science on policy, public understanding, and capacity-building by translating scientific evidence on urgency, irreversibility, and risk into actionable knowledge for decision-makers and society.** All Main Goals of OC5 need to run in parallel, though the focus of the first three-years is on *MG5.1 Policy Integration and Engagement,* with a summary of this MG presented at the first *Triennial Summit of the Decade of Action for Cryosphere Sciences*.

Mandate and Deliverables

The OC5 Working Group (WG5) is mandated to coordinate and oversee three interlinked streams of work:

MG5.1 Policy Integration and Engagement - Cryosphere Loss Prevention

- Strengthen the science–policy interface by integrating cryosphere-relevant metrics and concepts (e.g. irreversibility, thresholds, safe operating spaces) into formal policy and negotiation processes.
- Support targeted engagement with parliamentarians, negotiators, and ministries through tailored briefings and structured exchanges.
- Coordinate the production or endorsement of a policy-relevant "Cryosphere Science and Policy Report" every three years, aligned with the three-year Decade cycle.

MG5.2 Global Cryosphere Outreach and Communication

- Develop and implement a flexible Outreach Plan to increase the visibility, accessibility, and societal relevance of cryospheric science.
- Support bidirectional knowledge brokerage and audience-specific communication, including multilingual materials.
- Maintain readiness to respond to major scientific findings, political windows of opportunity, and misinformation or disinformation.

MG5.3 Education and Capacity Building

- Strengthen cryospheric scientific literacy across education levels and societal sectors.
- Support youth engagement, educator resources, and training of early-career and next-generation cryosphere scientists.
- Coordinate with existing education and research networks to avoid duplication and maximise impact.

Throughout the decade, this working group will report biannually to the Strategic Management Committee, on progress towards these goals. WG5 will report on the progressive application, across all MG's, of the globally recognized FAIR (Findable, Accessible, Interoperable, Reusable), CARE (Collective Benefit, Authority to Control, Responsibility, Ethics), and TRUST (Transparency, Responsibility, User Focus, Sustainability, Technology) principles within cryosphere knowledge, outreach and education products, applied ethically with consent, recognition and benefit-sharing.

Structure

a) Open and inclusive membership

WG5 is open to all interested experts and stakeholders contributing to the objectives of the Decade. General membership shall remain open to all applicants meeting the criteria of the Decade, with no fixed threshold or numerical cap, and all members shall be entitled to participate in the discussions and activities of the Group.

b) Leadership composition

The leadership of WG5 comprises:

- a Lead, appointed for an initial three-year term;
- three Associate Leads, appointed for an initial three-year term, jointly with the Lead, on the basis of complementarity of expertise, thematic coverage and geographical balance;
- Sub-task leads, who are in charge of their own teams
- 3 resource mobilization leads
- a Secretariat, hosted by an institution with the mandate, capacity and networks to support the scientific coordination of the Working Group.

The lead, associate leads and secretariat positions are nominated by the Ad Hoc Strategic Management Committee on the proposal of the UNESCO IHP Secretariat, following the open call for expressions of interest and considering scientific excellence, geographical and gender balance, interdisciplinary expertise, and diversity of background and generation. To promote the above principles, efforts will be made, where appropriate, to identify potential funding/support for individuals in leadership roles to participate.

c) Core group

A flexible core group shall be established to support coordination and continuity. The size and composition of the core group shall be determined by the Lead, Associate Leads and Secretariat, in consultation with the broader WG5 and the UNESCO Intergovernmental Hydrological Programme (IHP)Secretariat, on the basis of thematic needs and the level of expressed interest. The core group shall as a guiding principle include:

- a transdisciplinary mix of scientific, policy, practitioner and knowledge-holder profiles;
- geographical balance, with deliberate inclusion of expertise from polar regions, high mountains, snow and permafrost zones, and downstream / low-lying regions affected by cryosphere loss;
- dedicated seats for representatives of Indigenous peoples and local communities, in articulation with the Indigenous Advisory Committee of the Decade;
- dedicated seats for early-career scientists, as part of the next-generation capacity ambition of the Decade;
- expertise spanning cryospheric science, policy, science communication and education, together with knowledge brokers and outreach specialists able to reach diverse audiences; Participation in the Working Group is voluntary and non-remunerated

d) Roles and responsibilities

The responsibilities of the leadership positions are defined as follows:

- **The Lead** is responsible for the overall direction, scientific quality and delivery of WG5; represents the Group at the Strategic Management Committee and externally; chairs meetings;

ensures alignment with the Synthesis Document, with the Decade's Overarching Challenges and Main Goals, and with the Triennial reporting cycle.

- **The three Associate Leads** share thematic, geographical and outreach responsibilities; act jointly with the Lead on substantive decisions; ensure thematic completeness across the breadth of cryosphere science-policy, outreach and education; deputise for the Lead when required; and lead specific sub-streams of work as agreed within the Group.
- **The Secretariat** provides operational and institutional support to the Group; organises meetings and documentation; ensures interoperability with the host organisation's networks and existing programmes; liaises with the UNESCO IHP Secretariat and with the Secretariats of the other Working Groups to foster cross-cutting coordination among the Working Groups, the Cryosphere Finance Dialogue and the Alliance for the Cryosphere.
- **The sub-task leads and resource mobilization leads** provide thematic expertise and leadership specific to their assigned sub-tasks. These individuals will work closely with the associate leads to ensure delivery of assigned sub-task responsibilities and mobilization of outputs, where appropriate.

e) Term, rotation and continuity

Leadership and core-group positions are appointed for an initial three-year term, aligned with the three triennia of the Decade and the Triennial Cryosphere Summits. A transitional overlap of up to six months between outgoing and incoming officers shall be maintained to preserve continuity. Mid-term review by the Strategic Management Committee may lead to adjustments where required.

Modus Operandi

- The Working Group shall meet primarily online (tentatively 2 hours each month), with a frequency sufficient to maintain momentum, complemented by periodic in-person meetings aligned with major international events where feasible and provided funding resources are made available.
- The Working Group shall hold at least one joint meeting per year with the other Decade Working Groups (WG1 – WG4) to foster synergy, share progress on cross-cutting issues, align activities across Overarching Challenges, and avoid duplication of effort.
- Progress and outputs of the Working Group shall be reported biannually by the Core group to the Strategic Management Committee and showcased at the Triennial Cryosphere Summits. A Lead, two Associate Leads and a Secretariat will be nominated during the Committee's first meeting.
- Draft agendas and documents shall be circulated in advance of each meeting. Decisions of the Core Group and the Working Group shall be made by consensus.
- Members of the Strategic Management Committee shall be expected to consult and engage with their respective networks and constituencies, and to facilitate broad participation in the activities of the Working Groups. The Lead of the Working Group serves as a member of the Strategic Management Committee and maintains regular contact with the UNESCO IHP Secretariat.
- In line with the Synthesis Document, WG5 shall not endorse, fund, promote or otherwise legitimise geoengineering research or deployment, recognising the scientifically uncertain and potentially harmful nature of such interventions. Activities shall integrate Indigenous and local knowledge ethically, with consent, recognition and benefit-sharing, and shall comply with UNESCO rules and procedures.
- Articulation with other initiatives. WG5 shall actively articulate its work with the International Polar Year (IPY 2032-2033), the UN Decade of Ocean Science for Sustainable Development

(2021-2030), the International Decade of Sciences for Sustainable Development (2024-2033), and other relevant initiatives, to maximise synergies and avoid duplication.

Website and further details

Proceedings of the Decade, consultation summaries and deliverables will be published on the Decades official website: www.un-cryosphere.org